RAJALAKSHMI ENGINEERING COLLEGE

R Ashwin Kumar 251801033-2025-AIDS

R2

Started on	Wednesday, 26 August 2026, 10:53 PM
State	Finished
Completed on	Wednesday, 26 August 2026, 10:54 PM
Time taken	47 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    int x;
    scanf("%d", &x);

    int low = 0;
    int high = n - 1;
    int floor = -1;

    while (low <= high)
    {
        int mid = (low + high) / 2;

        if (arr[mid] == x)
        {
            floor = arr[mid];
            break;
        }
        else if (arr[mid] < x)
        {
            floor = arr[mid];
            low = mid + 1;
        }
        else
        {
            high = mid - 1;
        }
    }

    printf("%d\n", floor);

    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course